AIRTOURER MAINTENANCE MANUAL

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FOREWORD

Correct maintenance, repair and overhaul of a light aircraft ensures less time in the repair and overhaul shop, lengthens the aircraft life and renders satisfaction to the owner. Ease of maintenance has been a major consideration in the designing of the Airtourer.

The information presented within is divided into three sections: **Maintenance**, **Repair** and **Overhaul**.

The **Repair Section** contains instructions and descriptive information to enable permissible repairs to the aircraft structure to be carried out in the field. The decision of whether to repair or replace, in the case of a major repair, will be influenced by a comparison of the relative cost and time factors. The manufacturers should be consulted if there is any doubt about a repair not specifically mentioned within the repair section.

The **Maintenance Section** provides sufficient information to enable all normal servicing, inspection, trouble shooting, adjustments, removals and replacements to be carried out by an engineer in the field.

The **Overhaul Section** contains specific instructions to enable work to be carried out on units and assemblies removed from the aircraft, and is therefore primarily for the use of the approved repair organisation.

Amendment sheets will be issued as necessary and should be incorporated in the manual. Service Bulletins will also be issued from time to time to notify the owner of modifications, requirements and recommendations.

The **Airtourer Spare Parts Manual** is a supplementary publication to this manual.

ASSOCIATED INSTRUCTION MANUALS

Publication	Engine	Reference
Overhaul Manual	Continental O-200A	X 30011
Parts Catalogue	Continental O-200A	X 30081
Operators Handbook	Continental O-200A	TSD 2041
Overhaul Manual	Lycoming O-235-C	60294-7

Publication	Engine	Reference
Parts Catalogue	Lycoming O-235-C	PC 102
Operators Manual	Lycoming O-235-C	60297-9
Repair and Overhaul Manual	McCauley Propeller	651015

NOTES TO THE READER

> **WARNING** — Calls attention to methods, procedures or limits which must be followed precisely to avoid **injury or death** to persons.

> **CAUTION** — Calls attention to methods and procedures which must be followed to avoid **damage to equipment**.

> **NOTE** — Calls attention to methods which make the job easier.

Where components or items of equipment are replaced they must:

- Be covered by a Release Note, Affidavit or other certifying documentation
- Be installed by a person approved or licensed by the appropriate Airworthiness Authority

Overhaul of engine, engine accessories, instruments, electrical equipment, radio and propeller must be carried out by approved organisations.

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SUPPLEMENTS

- Airtourer T4, T5 & T6
- Airtourer T6/24

CHAPTER 1 — DIMENSIONS AND AREAS

AT.100

General

Parameter	Value
Wingspan	26' 0"
Length overall	20' 9⅜"
Height	7' approx.
Ground angle	4°

Wings

Parameter	Value
Incidence at root	3°
Dihedral (upper surface)	6°
Area	120 sq. ft.
Aspect ratio	5.7
Flapperon area total	18 sq. ft.
Aileron area total	8.68 sq. ft.

Empennage

Parameter	Value
Fin area	5.6 sq. ft.
Rudder area	6.5 sq. ft.
Rudder height	4' 0"
Fin and rudder offset	Nil
Elevator area	13.6 sq. ft.

Parameter	Value
Tailplane area	15.5 sq. ft.
Tailplane span	10' 10"
Tailplane incidence	0°
Tailplane dihedral	0° – 24' (measured at rib chord plane)

Fuselage

Parameter	Value
Width max.	3' 6"
Height (bottom to top of canopy)	4' 6"
Length (without engine and mount)	16' 8"

Landing Gear

Parameter	Value
Wheelbase	60.12"
Main gear track static	9' 6"
Tyre size	5 × 500

AT.115

General

Parameter	Value
Wingspan	26' 0"
Length overall	21' 6"
Height	7' approx.
Ground angle	4°

(Wings, Empennage, Fuselage and Landing Gear dimensions identical to AT.100 above)

CHAPTER 2 — LIFTING AND SHORING

1. General

When lifting the aircraft, **Slinging Beam Pt. No. 05009** and **Sling Pt. No. 05008** should be used in conjunction with the upper engine mount attachments and the rear fuselage adjacent to the fuselage steps.

2. Lifting Beam

The lifting beam consists of two back-to-back aluminium alloy extrusions. The fulcrum point can be varied to suit the aircraft state:

- Less wing
- Fully equipped
- With full fuel
- Less engine
- Less engine and wing

3. Sling

The rear sling is a 1" wide terylene strap capable of taking a load of 1,500 lbs. Pass the strap under the rear fuselage steps; two loops attach to the fuselage steps to prevent rearward sliding. The forward sling is a flexible wire rope assembly passed around the upper engine mount attachment clusters.

> **NOTE:** Fit the rear sling first. Ensure the forward part of the sling is hard up against the firewall, otherwise bending of the engine mount may result.

4. Lifting the Engine

Use engine sling **Pt. No. 05013/1** in conjunction with the lifting eye on the engine crankcase.

5. Raising the Nose Wheel

On aircraft Serial No. 66 and subsequent, a jacking mount is located at the bottom of the forward firewall face. For aircraft prior to Serial 66, depress the tailplane and secure with a rope through the fuselage tie-down eye.

> **CAUTION: DO NOT LIFT THE NOSE BY MEANS OF THE ENGINE LIFTING EYE.**

6. Jacking

Jacking mounts are situated on the under surface of the wing, aft of the main undercarriage legs. Remove the plug from the jacking mount, lift the aircraft and place a padded trestle under the fuselage behind the firewall.

CHAPTER 3 — LEVELLING AND WEIGHING

1. Levelling Points

The canopy sills immediately behind the windscreen are used for both lateral and longitudinal levelling. With the sills horizontal, the front face of the firewall is used as vertical datum.

2. Weight Summary

Item	AT.100 (lbs)	AT.115 (lbs)
Structure, including undercarriage	607	607
Power Plant	307.5	331.3
Fixed equipment	144.1	144.1
Empty weight, standard equipment	**1058.6**	**1082.4**
Pilot and passenger	340	340
Max. Fuel (29 galls IMP)	214	214
Oil (5 Qts IMP)	11.25	11.25
Baggage	100	100

3. Weighing the Aircraft

Level the aircraft on the canopy sills with scales under the main and nose wheels.

Example:

- Weight on main wheels: 818 lbs
- Weight on nose wheel: 247 lbs
- **Total weight: 1065 lbs**

4. C.G. Data

Example calculation (x = 34.9, y = 94):

Leading edge of S.M.C. is at station 70.9 (20.9" aft of firewall front face).

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C.G. Station = (818 × 94 + 247 × 34.9) / 1065 = 80.29
C.G. = (80.29 - 70.9) / 55.4 = 16.95% S.M.C.
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> **CAUTION: Always check calculations by measurement.**

5. Useful Load

Item	Wt. lbs	Arm (ins)	Moment (lb·ins)
Fuel (max.)	209	98	20,482
Oil	11.3	32	360
Pilot or passenger	170	90.2	15,385
Baggage	100	111	11,100

6. Optional Equipment

Item	Wt. lbs	Arm (ins)	Moment (lb·ins)
Bendix CNS 220 Transceiver	5.18	62	359.6
AWA Skyphone VHF Transceiver	6	62	372
HF Transceiver	11	62	682
Cabin Speaker	0.5	103	51.5
Artificial Horizon	2.8	64	179.2
Directional Gyro	2.6	64.5	167.7
Rate of Climb Indicator	1.0	64.5	64.5
Carb. Temp. & Bulb Adaptor	0.8	65.00	52.0
Cyl. Head Temp. Gauge + Thermocouple	1.2	66.5	79.7
Fuel Pressure Gauge & Pipes	0.7	66.5	46.6

Item	Wt. lbs	Arm (ins)	Moment (lb·ins)
Clock	0.4	66.5	26.6
Outside Temp. Gauge	0.1	69.00	6.9
Vacuum Gauge	0.4	66.5	26.6
Second Venturi	2	77	154
Ammeter	0.3	69.00	20.7
Rotating Beacon	2.1	143	300
Landing Light	1.0	21	21

7. C.G. and Weight Limits

Maximum aerobatic weight: 1,550 lbs
Maximum normal category weight: 1,650 lbs

C.G. Limits — Airtourer 100

- **Forward limit:** 82.8" aft of datum at 1,650 lb; 79.2" at 1,330 lb or less (linear variation between)
- **Aft limit (Normal):** 86.35" aft of datum at all weights
- **Aft limit (Aerobatic):** 84.9" aft of datum at all weights

C.G. Limits — Airtourer 115

- **Forward limit:** 82.25" aft of datum at 1,650 lb; 78.75" at 1,275 lb or less (linear variation between)
- **Aft limit (Normal):** 86.35" aft of datum at all weights
- **Aft limit (Aerobatic):** 84.9" aft of datum at all weights

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CHAPTER 4 — TOWING AND TAXYING

1. Towing and Pushing

Insert tow bar **02-05006** into the hollow axle of the nosewheel. With the propeller horizontal, grasp it at the root to push or pull.

> **CAUTION:** Do not push or pull at the outboard end of the propeller. Nosewheel travel is limited to approximately 12½° port and 16½° starboard.

2. Taxying

Release the parking brake and use minimum power to start moving (~1,200 RPM). During taxi and especially downwind, hold RPM down to prevent excessive speed. Use brakes sparingly.

> **CAUTION:** Care should be exercised when taxying over rough fields. Taxying over loose gravel should be done at approximately 1,000 RPM.

CHAPTER 5 — PARKING AND MOORING

1. Parking

To apply parking brakes:

1. Pull the brake handle fully back
2. While holding pressure, pull out the parking brake knob on the centre console
3. Release handle

To release: Pull brake handle.

2. Covers

Cockpit enclosure and engine covers should be kept in position when parked in the open.

3. Mooring

Three mooring eyes are provided: one at each wing tip and one at the rear of the fuselage. Use dog clips to attach mooring ropes.

Gust lock installation:

1. Fit the metal bracket over the top of the control column
2. Loop the cable ends over the clips on the top longeron forward of the windscreen arch

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CHAPTER 7 — SERVICING

1. Refuelling

Aircraft is serviced with **80 octane minimum**. Tank capacity: 29 gallons (including 0.3 gallon unusable).

1. Ensure purity of fuel

2. Bond fuel filling hose to aircraft before refuelling
3. Do not overfill
4. Drain sufficient fuel from the tank drain to clear water

2. Lubrication

Airtourer 100

Continental Specification MHS24

Ambient Temp.	Grade
Below 40°F	Aviation Grade 1040 – SAE 20
Above 40°F	Aviation Grade 1080 – SAE 40

Airtourer 115

Lycoming Specification 30-1-E

Condition	Straight Mineral	Multi-Grade Additive	Oil Inlet Temp Desired	Max
Above 60°F	SAE 50	SAE 40 or 50	82°C	118°C
30°–90°F	SAE 40	SAE 40	82°C	118°C
0°–70°F	SAE 30	SAE 40	76°C	107°C

3. Filling Oil Sump

Service with specified grade oil. Normal sump capacity: **5 Imp. quarts (6 U.S. quarts)**.

4. Cleaning

- **External surfaces:** Detergent wash, then high-grade automotive polish
- **Painted surfaces:** Mild soap and water, then automotive wax
- **Engine compartment:** Suitable solvent, then dry thoroughly
- **Upholstery:** Vacuum cleaner; foam detergent for stains; avoid saturating with volatile solvent
- **Propeller:** Wipe occasionally with an oily rag
- **Windshield/Canopy:** Household detergent and warm water; never dry-wipe

5. Topping Up Nose Leg Strut

With nose leg depressed until steering arm is 4¼" below the upper bearing, top up with approved shock absorber oil until excess spills from filler hole. Bounce and refill until level remains constant.

6. Topping Up Brake System

Unscrew cap on master brake cylinder and top up with specified hydraulic fluid.

> **CAUTION:** Do not mix different specifications of hydraulic fluid.

7. Bleeding the Brake System

The brake system must be bled whenever a brake line has been disconnected. Use a hydraulic hand pump with correct grade of brake fluid. Continue pumping until fluid flowing into the jar is free of air bubbles.

> **CAUTION:** Do not replenish with fluid drained from the system.

8. Fuel Filter Check & Renewal

Check every **25 hours**. Unscrew the central bolt, lower the element canister and bowl, clean and renew element if clogged.

9. Topping Up the Battery

Check every **week or 25 hours** (whichever comes first). Top up with distilled water to the level of the perforated plate.

10. Bleeding the Fuel System

Bleed whenever a fuel line has been disconnected. With fuel in tank, Master Switch ON, electric fuel pump ON — loosen the inlet hose at the carburettor end until fuel flow is free of air bubbles.

11. Cleaning the Air Filter

Clean every **100 hours** or more often in dirty conditions. Wash in clean unleaded petrol, blow from inside to outside, then spray or brush a light coating of engine oil on front and rear element surfaces.

> **CAUTION: DO NOT OVERTIGHTEN FILTER LOCKING STUDS.**

CHAPTER 8 — INSPECTION

1. Daily Inspection

Carried out prior to the first flight of each day.

2. Between Flight Inspection

1. Inspect wings, fuselage, empennage and control surfaces for security and obvious damage
2. Ensure windshield and canopy are clean
3. Fuel tank serviced and filler cap secure; fuel vent clear
4. Visually check underside of fuselage for fuel leaks

3. 25 Hourly Inspection

(Every 25 hours or once a month, whichever comes first)

1. Check filter bowl for water and sediment
2. Inspect aileron differential control boxes
3. Check battery level and top up if necessary
4. Drain sump and refill with specified engine oil
5. Clean main fuel filter
6. Lubricate aileron and flapperon pivot joints

4. 100 Hourly Inspection

(Every 100 flying hours — includes all daily inspection requirements)

Airframe

1. Check flight and serviceability log
2. Clean aircraft internally and externally
3. Inspect undercarriage legs, wheels, tyres, nose wheel strut
    - Check tyre inflation: **Main 23 psi / Nose 12 psi (AT100) or 16 psi (AT115)**
4. Inspect cabin interior, seats, safety belts, canopy and windscreen
5. Inspect fuselage skin for damage, dents, loose rivets, corrosion
6. Inspect empennage (tailplane, elevators, fin, rudder)
7. Inspect wing skin, ailerons, flapperons, centre flap, gap tapes
8. Inspect all flying controls for full and free travel

9. Lubricate per lubrication chart

Instruments

1. Check flight and serviceability log
2. Inspect instruments for security, condition and legibility
3. Check compass needle friction; altimeter; gyro caging devices
4. Drain pitot-static watertraps; check orifices; carry out leak test
5. Check venturi/vacuum system

Electrical

1. Check battery electrolyte level and specific gravity; inspect battery box for corrosion
2. Inspect starter, generator, commutators, brush holders
3. Check all switches, circuit breakers, generator warning light
4. Check stall warning system function

Engine

1. Clean engine; drain oil; remove and inspect oil filters for metal particles
2. Inspect engine mount bolts and rubbers
3. Inspect cylinders, valve gear, rocker covers
4. Remove spark plugs; check and clean; reset gaps:
    - **Continental O-200-A:** 0.015"–0.018"
    - **Lycoming O-235-C1:** 0.025" (unshielded), 0.015" (shielded)
5. Check cylinder compressions
6. Inspect magnetos and ignition wiring; check point gaps (0.012"–0.024")
7. Inspect cylinder baffles
8. Inspect fuel system plumbing, filters, tank vent
9. Inspect and service carburettor, hot air system, air filter
10. Inspect induction and exhaust manifolds, heat exchange shrouds
11. Inspect engine controls (throttle, mixture, carb heat)
12. Inspect cowlings and fasteners
13. Inspect propeller and spinner; check track within **0.125"**
14. Test run engine per Chapter 9

5. Special Inspections

Heavy or Overweight Landing

1. Inspect complete airframe externally for visible damage (sheared rivets, buckled skin)
2. Inspect mainplane-to-fuselage attachments
3. Inspect empennage structure
4. Inspect engine mount attachments and adjacent structure
5. Remove undercarriage fairings; inspect leg attachments (magnetic particle inspection if doubt exists)
6. Inspect nose leg strut bearings and mounting structure
7. Inspect wheels, axles, bearings and brake mechanisms
8. Functional test of instruments and radio; check battery

Severe Turbulence

1. Inspect complete airframe externally
2. Inspect mainplane-to-fuselage and empennage attachments
3. Check engine mountings
4. Inspect control surfaces, hinge points, cables and bellcranks
5. Check seats, seatbelts, instruments and equipment

Lightning Strike

1. Detailed external inspection for burning, scorching and pitting (especially aerials, wing extremities)
2. Check control surface hinge points and bonding
3. Check all internal bonding; inspect fuel lines and vents
4. Functional check of all electrical, instrument and radio services
5. Carry out compass swing

Propeller Damage (Ground/Obstruction Contact)

1. Remove propeller for inspection by approved organisation
2. Check engine crankshaft run-out at propeller flange — **maximum 0.005" (full indicator reading)**
3. Inspect crankcase in vicinity of engine mounting feet and crankshaft thrust bearing housing

4. Inspect engine mount frames for bowing, weld condition, and attachment bolts for shear

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CHAPTER 9 — GROUND RUN-UP PROCEDURE

Continental O-200-A Engine (Airtourer 100)

Pre-Starting

- Ignition switch: **OFF**
- Mixture: **FULL LEAN**
- Turn propeller over by hand 4 revolutions, noting compressions and impulse operations

Starting Procedure

Item	Action
Fuel	ON
Master Switch	ON
Electric Pump	ON
Generator Switch	ON
Prime	Two strokes of throttle
Throttle	Slightly open (~½" forward)
Mixture	FULL RICH
Carburettor Heat	COLD
Ignition Switch	BOTH ON
Starter	Establish "All Clear" then PULL

> **CAUTION:** Limit starter operation to 10 seconds at 2-minute intervals. If no oil pressure within 30 seconds, stop and investigate.

Warm Up

Idle at **800 RPM** for approximately 3 minutes (or until oil temperature minimum 24°C). Check magneto switches for dead cut. Check carburettor heat operation.

Run-Up

Increase smoothly to **1,700 RPM**. Check magneto drop — maximum **100 RPM** on either magneto; maximum difference **50 RPM**.

Operating Limits

Parameter	Value
Oil Pressure (normal)	30–60 PSI
Oil Pressure (idling min.)	10 PSI
Cylinder Head Temp. max.	274°C (525°F)
Oil Temperature max.	107°C (225°F)
Fuel Pressure	2–4 PSI

Engine Stop

Run at 800–1,000 RPM to cool. Then: Mixture → FULL LEAN. After engine stops: Ignition OFF, Throttle CLOSED, Electric Pump OFF, Generator OFF, Master Switch OFF.

Lycoming O-235-C2 Engine (Airtourer 115)

Starting Procedure

Item	Action
Fuel	ON
Master Switch	ON
Electric Fuel Pump	ON
Generator Switch	ON
Prime	3 strokes of priming pump (or 3 throttle strokes if no primer)
Throttle	Slightly open (~½" forward)
Mixture	FULL RICH
Carburettor Heat	COLD

Item	Action
Ignition Switch	BOTH ON
Starter	Establish ALL CLEAR then PUSH

Operating Limits

Parameter	Value
Oil Pressure (normal)	65–85 PSI
Oil Pressure (idling min.)	25 PSI
Cylinder Head Temp. max.	260°C
Oil Temperature max.	118°C
Fuel Pressure	2–5 PSI

CHAPTER 11 — ELECTRICAL SYSTEMS

1. Electric Power Supply System

12 volt DC, single wire earth return system comprising: battery, generator, voltage/current regulator, starter, battery master relay, and electrical panel with generator warning light, circuit breakers and switches.

2. Battery

12 volt, 24 ampere-hour capacity, mounted in battery box on the firewall. Isolated by battery relay controlled by master switch.

3. To Remove Battery

1. Remove two 4BA screws locking battery cradle
2. Slide battery and cradle from box
3. Disconnect **earth (negative) cable first**
4. Disconnect positive cable
5. Raise forward end of cradle over stop and withdraw

> **CAUTION: Always remove the earth cable first and replace it last.**

5. Battery Maintenance — Hydrometer Readings

Specific Gravity	State of Charge
1.280	100% charged
1.250	75% charged
1.220	50% charged
1.190	25% charged
1.160	Practically dead

(Readings based on electrolyte temperature of 80°F)

6. Battery Fault/Remedy

Fault	Probable Cause	Remedy
No power to bus when master on	Dead battery	Replace or charge battery
No power to bus when master on	Defective master switch or wiring	Repair wiring / replace relay
No power to bus when master on	Faulty battery cable	Replace cable; clean and reconnect
Supplies power but won't turn engine	Low battery	Charge battery
Supplies power but won't turn engine	Faulty battery cables	Clean and reconnect
Supplies power but won't turn engine	Cell shorting under load	Replace battery
Supplies power but won't turn engine	Defective starter	Repair wiring / replace starter
Excessive water usage	Charging rate too high	Adjust or replace regulator

8. Battery Box Maintenance

Clean periodically with strong bicarbonate of soda solution. Remove hard deposits with wire brush. Flush with clean water and check drain tube is clear.

> **WARNING:** Do not allow acid deposits to contact skin or clothing. Wash immediately with soap and water.

11. Generator Power System

20 ampere generator, voltage regulator, generator field switch, isolation circuit breaker and warning light. Charging rate should be 2–5 amps with normal electrical load.

22. Stall Warning

Audible warning circuit: lift detector switch in leading edge of port wing, wired to warning horn under luggage compartment floor. Activates when stalling is imminent with flaps down.

CHAPTER 13 — FLIGHT CONTROLS

1. General

The flight control system consists of aileron/flapperon, flap, rudder, elevator and elevator trim control systems. All can be operated from either side except the flap control (port side only).

3. Adjustment of Aileron and Flapperon Controls

1. Set control stick vertical (aircraft in rigging position)
2. Adjust flapperon push-pull rods until flapperons are in neutral position
3. Check ailerons are in neutral
4. Adjust aileron push-pull rod in differential box if needed (yellow band identifies pinned end)
5. Set flapperon/aileron stops: **23° to port and 23° to starboard**

Angular movement checks (stick held 23° to port):

Surface	Movement
Port flapperon	7° 54' up
Starboard flapperon	4° 25' down
Port aileron	8° 10' up
Starboard aileron	3° 3' down

Flapperon Neutral Setting

The flapperons are in neutral when the trailing edge centreline is **22.5" below the fuselage datum line**.

10. Rudder Control System

Stirrup-type rudder pedals connected via push-pull rod to idler lever, with pre-stretched cables to the rudder. Rudder movement limited by a stop at the base of the fin.

Adjustment

1. Set port pedals in rearmost notch; adjust push-pull rod until idler lever is square with aircraft centreline
2. Adjust cable turnbuckles to centre the rudder (horn coincides with top of fin)
3. Tension cables to **50 lbs**

14. Elevator Control System

Control column linked to elevator via two push-pull rods joined at rear of cockpit by idler lever.

Control limits:

- AT100: 12½° forward; 14° 12' aft
- AT115: 12½° forward; 18° 23' aft

Elevator travel:

- AT100: 19° up / 15° down
- AT115: 22° up / 15° down

CHAPTER 14 — FUEL SYSTEM

1. General

Fuel drawn from tank via fuel cock → filter → tee fitting → electric pump or engine driven pump → carburettor.

12. Fuel Tank

Rubber cell type, contained between inboard wing ribs. Usable capacity: **28.7 Imperial gallons**; total capacity: **29 Imperial gallons**.

18. Fuel System Trouble Chart

Fault	Probable Cause	Check & Remedy
No fuel in carburettor	Fuel cock not on	Turn on
No fuel in carburettor	Fuel line disconnected	Inspect and reconnect
No fuel in carburettor	Fuel tank empty	Service with correct grade
No fuel in carburettor	Carburettor filter blocked	Inspect and clean
No fuel in carburettor	Electric pump faulty	Check pump operation
Fuel starvation after starting	Broken/loose fuel lines	Inspect and tighten/replace
Fuel starvation after starting	Blocked fuel vent	Inspect and clean
No fuel when electric pump operated	Defective pump switch	Check switch continuity
No fuel when electric pump operated	Defective electric pump	Listen for clicking; if no flow, replace
No fuel quantity indication	Circuit breaker open	Check and replace if defective
No fuel quantity indication	Defective indicator or transmitter	Test by swapping indicator

CHAPTER 16 — LANDING GEAR

1. General

Main landing gear: two tapered spring steel cantilever legs clamped and bolted to secondary structure in wing. 5" diameter Dunlop wheel assemblies with hydraulic disc brakes.

3. Installation of Main Landing Gear — Torque Values

- Clamp plate bolts (AN6-H16A): **196–240 in·lbs**
- Inner attachment bolt (Pt. No. 61033/1): **456–504 in·lbs**

7. Main Wheel Alignment

Toe-in should be **1/16" maximum**. Check with aircraft empty and no fuel.

11. Brakes System

Dunlop hydraulic disc brakes on each main wheel, controlled by two brake handles. Parking brake valve controlled by Bowden cable terminating in cockpit knob.

CHAPTER 42 — ENGINE DATA

Engine Data

Parameter	AT100 — Continental O-200-A	AT115 — Lycoming O-235-C2
BHP rated max	100 at 2,750 RPM	115 at 2,800 RPM
Displacement	200.91 cu. in.	233.3 cu. in.
Bore	4.0625"	4.375"
Stroke	3.875"	3.875"
Compression ratio	7.0:1	6.75:1
Oil sump capacity	4.2 Imp. qts (6 US qts)	4.2 Imp. qts
Valve mechanism	Hydraulic non-adjustable	Adjustable tappets
Valve clearance (cold)	0.030–0.110"	0.010"
Oil pressure (normal)	30–60 PSI	65–85 PSI
Oil pressure (idling min.)	10 PSI	25 PSI
Min. oil temp. for takeoff	75°F (24°C)	104°F (40°C)
Max. oil temp.	225°F (107°C)	245°F (118°C)
Cylinder head temp. max.	525°F (274°C)	500°F (260°C)
Carburettor	Marvel-Schebler MA-3SPA	Marvel-Schebler MA-3A
Magnetos	Bendix Scintilla S4LN-21 (×2)	S4LN-20 (RH), S4LN-21 (LH)
Firing order	1-3-2-4	1-3-2-4
Spark timing (°BTC)	28°	25°
Spark plug	Champion EM42E(C)	Champion EM42E(C)

Parameter	AT100 — Continental O-200-A	AT115 — Lycoming O-235-C2
Spark plug gap	0.018" shielded	0.015"

CHAPTER 50 — GENERAL REPAIR PROCEDURES

2. Basic Rules for Repair

- **Assessment:** Examine surrounding parts methodically; check bolt/rivet holes for elongation; perform alignment checks where major damage occurs
- **Support:** Support structure on suitable trestles before commencing repairs
- **Strength:** Use materials of quality and specification at least equal to original; use same type and size of rivet

Preparation

- **Dents:** Dress with fibre mallet and wooden blocks; examine for cracks
- **Nicks:** Clean to smooth-edged curvature
- **Tears:** Dress to regular shape; remove ragged edges
- **Scores:** Negligible if depth ⩽10% of plate thickness and length <15% of panel width
- **Cracks:** Drill No. 40 (0.098") hole at each extremity to prevent spreading
- **Corrosion:** Clean out and treat with Alodine if within score/scratch limits

3. Skin Panel Damage Assessment

Damage Type	Location	Assessment
Shallow dent (no cracks/creases)	Less critical	Negligible if ⩽0.2" deep, <5" diameter (limit: one per panel)
Shallow dent (no cracks/creases)	High critical	Negligible if ⩽0.06" deep, <2" diameter
Sharp dent	All panels	Replace damaged area
Dent with crease	All panels	Replace damaged area
Holes	Less critical	Patch if ⩽1" diameter
Holes	Semi-critical	Patch if ⩽¾" diameter

Damage Type	Location	Assessment
Cracks up to 15% panel width	Less critical	Drill 1/8" hole at crack end and patch
Cracks	Semi/high critical	Replace cracked area

6. Rivet Spacing

Rivet Size	Minimum Spacing	Maximum Spacing
3/32"	0.50"	0.70"
1/8"	0.50"	1.00"
5/32"	0.60"	1.00"

Minimum edge distance: **2D** (where D = rivet diameter).

7. Rivet Length

```
Rivet length = material thickness + (1.5 × rivet diameter)
```

8. Rivet Types

> **WARNING:** Fit the same type of rivets originally used. Tucker Pop rivets must not be substituted for AN solid or Avdel rivets.

Monel Tucker Pop Rivets (Table 2)

Rivet Dia.	Code	Length	Domed Head Total Thickness	Csk Head Total Thickness
1/8"	419	0.19"	0.020–0.070	0.030–0.100
1/8"	424	0.24"	0.071–0.120	0.101–0.150
1/8"	429	0.29"	0.121–0.170	0.151–0.200
1/8"	435	0.35"	0.171–0.230	0.201–0.260
5/32"	524	0.24"	0.051–0.100	0.081–0.130
5/32"	530	0.30"	0.101–0.160	0.131–0.190
5/32"	537	0.37"	0.161–0.230	0.191–0.260

Coding: TLP = Tucker pop-monel; K = Countersunk 120°; D = Dome head; BS = Break stem mandrel

AN Solid Rivets (Table 4)

- `AN426AD` = 100° csk head, A17ST aluminium alloy
- `AN470AD` = round head, A17ST aluminium alloy
- First dash number = diameter in 1/32nds
- Second dash number = length in 1/16ths

CHAPTER 51 — STANDARD REPAIRS

2. Fibre Glass Repairs

Use any commercial polyester resin. For cracks: apply resin and one/two layers of glass cloth inside as a doubler. For holes: trim damaged area, apply glass mat over hole inside, cover outside with cellophane, apply epoxy resin and sand smooth after curing.

4. Repairs to Acrylic Canopy

- **Cracks:** Drill 1/8" holes at each end; scrape to "V" shape; fill with acrylic cement
- **Holes:** Clean to circular shape; bevel edges to 45°; fit acrylic patch of same thickness; secure with acrylic cement
- **Scratches:** Sand with No. 320 (or finer) wet sandpaper using circular motion; buff with motor-driven pad at max. ~2,000 ft/min to restore transparency

> **NOTE:** Repair only where pilot's field of view is not affected. Damaged areas affecting vision require panel renewal.

> **CAUTION: Never use coarser than No. 320 abrasive.**

7. Repairs to Welded Joints

Use low carbon content welding rod. Grind or file joint to "V" profile for maximum penetration.

Welding rod composition:

Element	Limit
Carbon	0.10% max
Manganese	0.30%
Phosphorous	0.04% max
Sulphur	0.04% max

SUPPLEMENT — AIRTOURER T4, T5 AND T6

Associated Instruction Manuals

Publication	Reference
Overhaul Manual — Lycoming O-320	60294-7
Parts Manual — Lycoming O-320	PC103
Operators Manual — Lycoming O-320	60297-16
Owners Manual — Hartzell	115
Repair Manual — Sensenich	N.P.N

Dimensions

Parameter	Value
Wingspan	26' 0"
Length overall	22' 0"
Height	7' approx.
Wing area	120 sq. ft.
Tyre size	T4/T5: 5×500; T6: 5×600

Weight Summary

Item	T4/T5 (lbs)	T6 (lbs)
Structure, including undercarriage	675	700
Power Plant	358	358
Fixed equipment	142	182
Empty weight	**1,175**	**1,250**
Max. aerobatic weight	1,650	1,800
Max. normal category weight	1,750	1,900

C.G. and Weight Limits

T4/T5

- **Forward limit:** 81.5" at 1,750 lb; 79.2" at 1,400 lb or less (linear variation between)
- **Aft limit (Normal & Aerobatic):** 86.0" at all weights

T6

- **Forward limit:** 81.0" at 1,450 lb or less; 84.0" at 1,800–1,900 lb (linear variation)
- **Aft limit (Normal):** 86.0" at all weights
- **Aft limit (Aerobatic):** 86.1" at all weights

Datum: 50" forward of the vertical face of the firewall

Engine Data — Airtourer 150 HP (O-320-E1A/E2A)

Parameter	Value
Engine	Lycoming O-320-E1A or E2A
BHP rated max	150 at 2,700 RPM
Displacement	319.8 cu. in.
Bore	5.125"
Stroke	3.875"
Compression ratio	7.00:1

Parameter	Value
Oil sump capacity	6.4 Imp. qts
Valve mechanism	Hydraulic
Valve clearance (tappets collapsed)	0.020"–0.080"
Oil pressure (normal)	60–90 PSI
Oil pressure (max starting/warm-up)	100 PSI
Oil pressure (idling min.)	25 PSI
Min. oil temp. for takeoff	104°F (40°C)
Max. oil temp.	245°F (118°C)
Cylinder head temp. max.	500°F (260°C)
Carburettor	Marvel-Schebler MA-4SPA
Magnetos	Bendix Scintilla S4LN-20 (RH), S4LN-21 (LH)
Firing order	1-3-2-4
Spark timing (°BTC)	25°
Alternator	Prestolite ALE 8406, 40A at 12V
Starter	Prestolite MZ4206, 12V
Spark plug	Champion EM42E

Electrical System (T4/T5/T6)

12V DC single wire earth return; 40A Prestolite alternator; battery in rear fuselage (port side).

Ground Run-Up — O-320 Engine

Follow Chapter 9 procedure with the following changes:

- Oil pressure: 60–90 PSI normal; 100 PSI max (starting/warm-up)

Airtourer T5/T6 — additional steps at 1,800 RPM:

1. Move constant speed control fully rearward; note RPM drop
2. Return to full fine (forward); confirm RPM returns to original

3. Check maximum RPM: **2,700**

SUPPLEMENT — AIRTOURER T6/24

Differs from T6 in electrical and instrument installations only.

Electrical System

24V DC single wire earth return system.

Battery: 24V nickel-cadmium, 13 Ah capacity, mounted in rear fuselage (port side).

Alternator: Prestolite ALT 8404, 50A at 28V.

Voltage Regulator: Prestolite VSF-7400 solid state; adjusted to 28.4V under 10–15A load.

Ammeter: Centre reading 50-0-50 type, connected in series with main bus bar to battery relay.

Starter: Prestolite MHB4001, 24V.

Avionics

Bendix RT-221 VHF transceiver (117.00–135.95 MHz) or Narco Mk VHF/VOR system; powered from 10A circuit breaker at 28V DC. 17" whip antenna on port side of rear fuselage.

Instruments

Standard: Altimeter, ASI, Turn & Balance, Rate of Climb, Artificial Horizon, Directional Gyro, Tachometer, Triple oil/temp/fuel pressure gauge, Manifold pressure gauge, Compass, Fuel contents gauge, Ammeter, Vacuum indicator

Optional: Carb inlet temp, cylinder head temp, outside air temp, flap position indicator, elevator trim indicator

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End of Airtourer Maintenance Manual — Reprint March 2007